



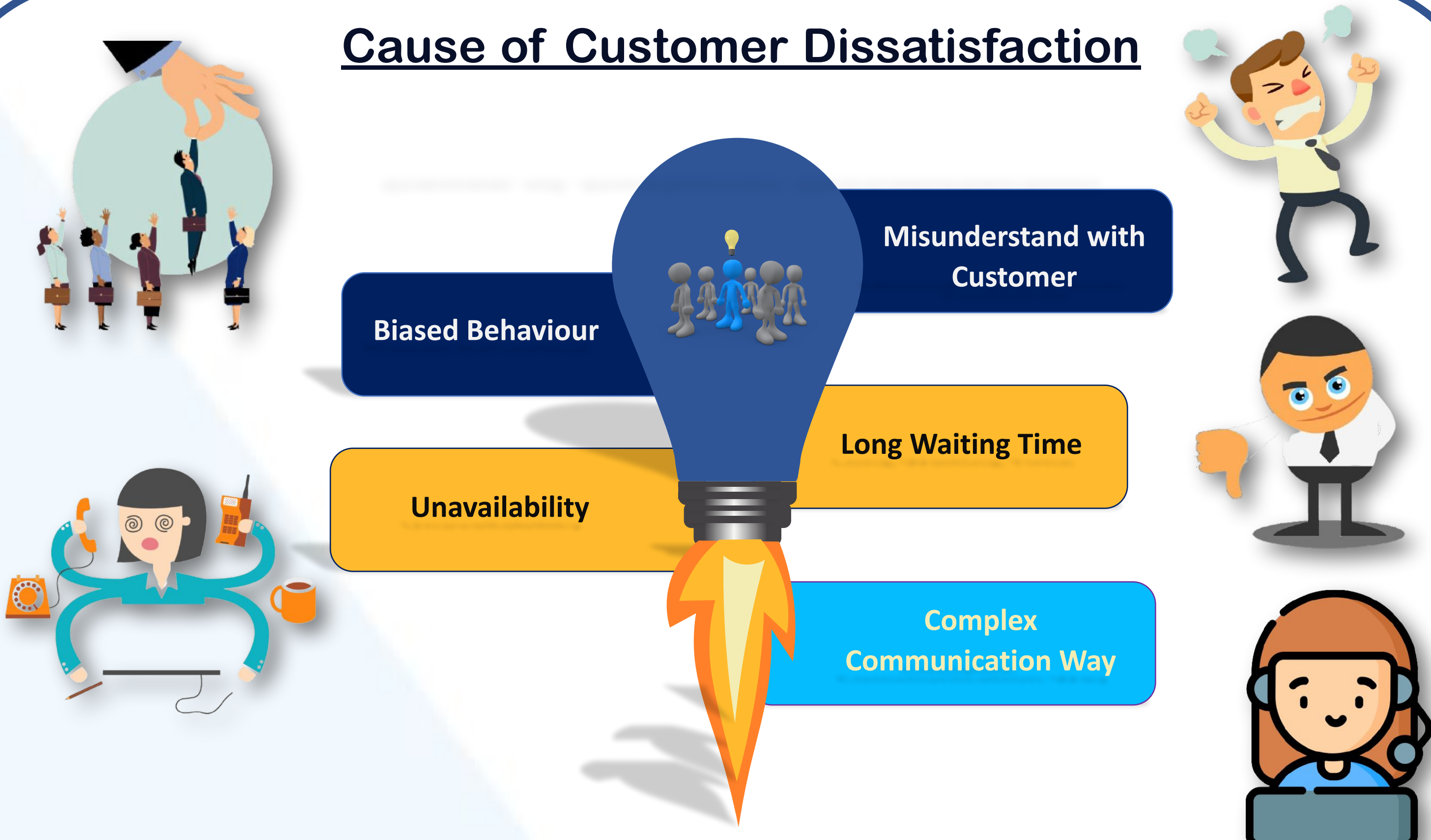
AI RECEPTION

AI Reception: Human-Robot Interaction in Bengali Language for Better Customer Satisfaction

Rahad Arman Nabid, Shehan Irteza Pranto, Ahnaf Mozib Samin, Nabeel Mohammed, Mohammad Nurul Huda, Farhana Sarker, Khondaker A. Mamun



Cause of Customer Dissatisfaction



Domains of AI Reception



01 Educational Institution

Guardian/Students can collect results and admission information

02 Hospital

Patients can reserve doctor's appointment & telemedicine facilities

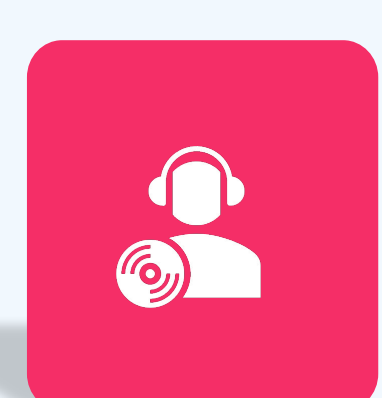
03 Bank

Customers can get accounts information & FAQ

04 Office

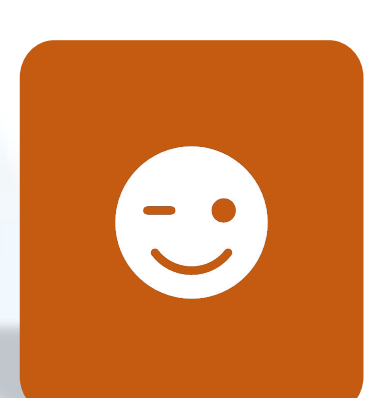
Service user can get correct information & unbiased response

Features of AI Reception



G-governance

Automation of reception management.



Customer Satisfaction

Unbiased response increase customer satisfaction.



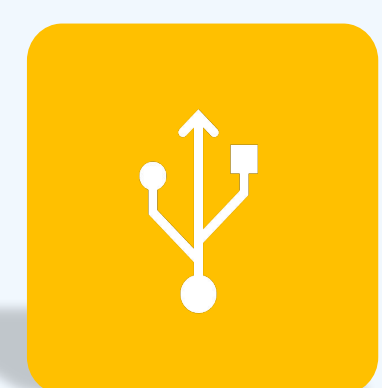
Covid Safety

Reduce mess congregation.



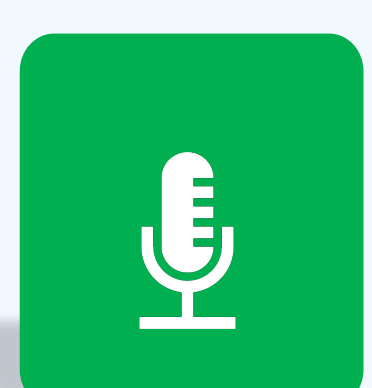
Offline & Online Integration

Low cost offline and online integration.



Easy Access of Information

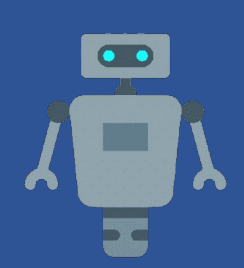
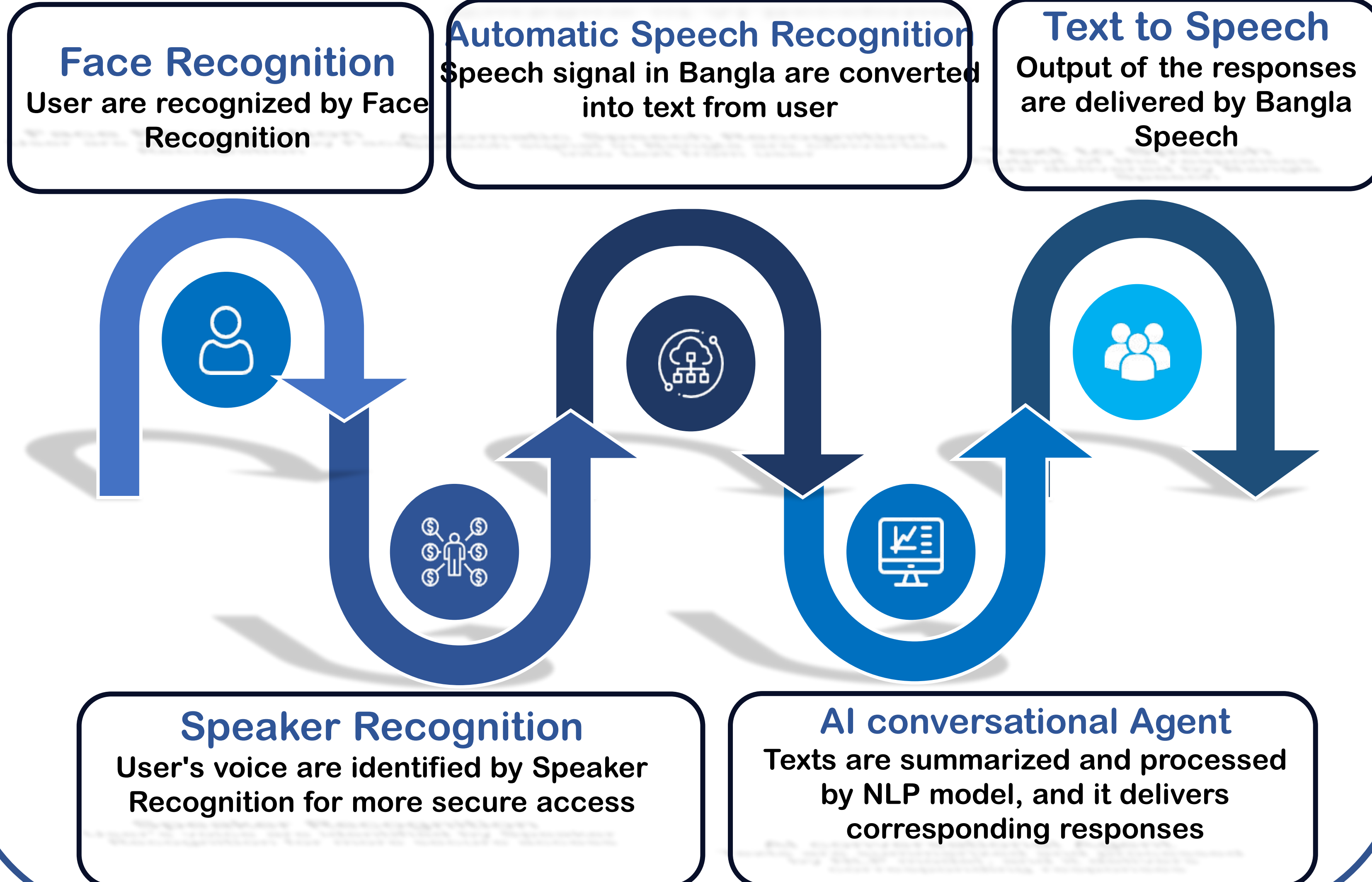
Easy access of correct information.



Universalize Bangla Language

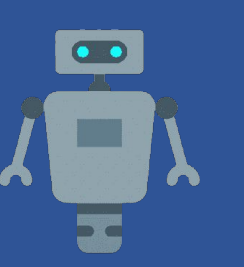
Communicating through Bangla Language for general people.

Workflow of AI Reception



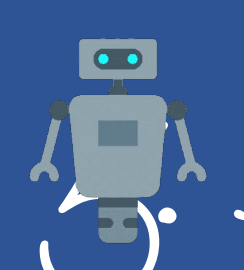
1. Face Recognition

- Face Recognition is the first step of authentication.
- It captures & recognizes users' images.
- After saving the data it allows to go the next step.



3. Automatic Speech Recognition

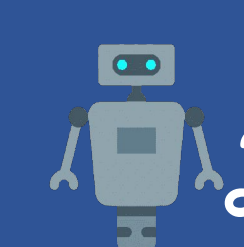
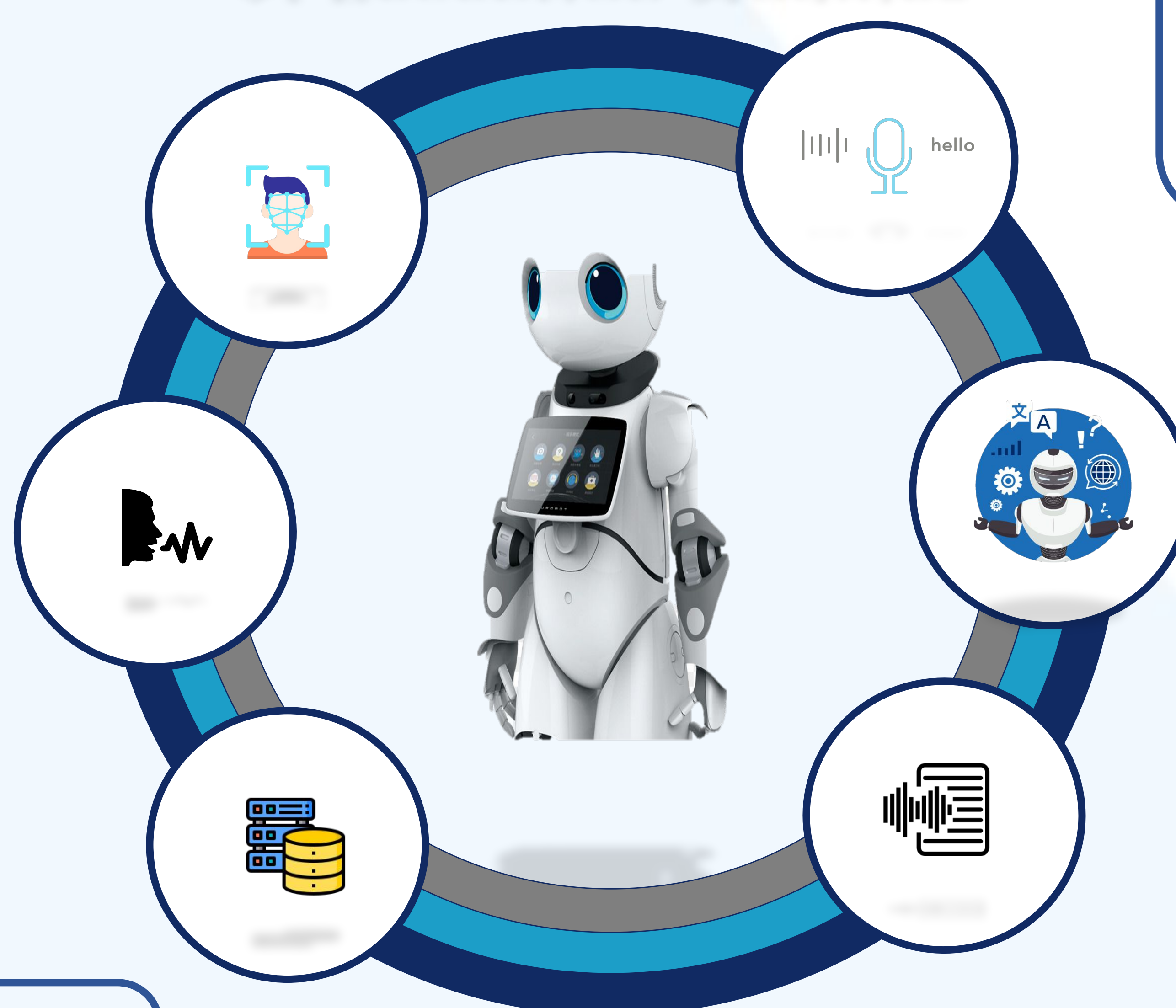
- It records the queries of the user in speech format
- In the next step speech signal is converted into text format



Database Interaction

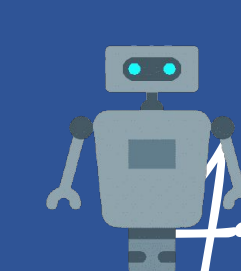
- Domain specific questions and answers are arranged into the database.
- Response of the user's data will be saved for future usages.

AI Reception Modules



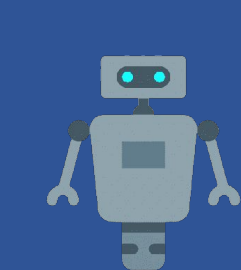
2. Speaker Recognition

- Speaker recognition is second step of the authentication.
- It records the users voice & extracts the features with GMM model
- After successful recognition it allows to activate the conversational agent with welcome message.



4. Conversational Agent

- Queries in text format from ASR will summarized and encoded with sentence transformer.
- Encoded text will be stored in database



6. Text to Speech(TTS)

- Answers to the question will be transformed into speech format with this model.

PUBLICATIONS

- [1] Rahad Arman Nabid, Shehan Irteza Pranto, Nabeel Mohammed, Farhana Sarker, Mohammad Nurul Huda, and Khondaker A. Mamun. "AI Reception: An Intelligent Bengali Receptionist System Integrating with Face, Speech, and Interaction Recognition" In 2021 Int'l Conference on Bangabandhu and Digital Bangladesh (ICBDB), 03-04 December 2021. (Accepted)
- [2] Shehan Irteza Pranto, Rahad Arman Nabid, Ahnaf Mozib Samin, Nabeel Mohammed Farhana Sarker, Mohammad Nurul Huda, and Khondaker A. Mamun. "HumanRobot Interaction in Bengali language for Healthcare Automation integrated with Speaker Recognition and Artificial Conversational Agent" In 2021 3rd International Conference on Electrical and Electronic Engineering (ICEEE), IEEE, 22-24 December 2021.(Manuscript Submitted)
- [3] Shehan Irteza Pranto, Rahad Arman Nabid , Ahnaf Mozib Samin , Nabeel Mohammed, Farhana Sarker , Mohammad Nurul Huda, and Khondaker A. Mamun. "Intelligent Call Center Support integrating with Speaker and Interaction Recognition using NLP", In 2nd International Conference on Emerging Technology in Computing, Communication and Electronics (ETCCE 2021), 21-23 December 2022 (Manuscript Submitted)

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Collaboration



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